

Vol 4 Chapter 12 — The Observational Reanalysis Program (SP-27)

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Chapter 12 — The Observational Reanalysis Program (SP-27)

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

This is the closing chapter of Volume 4. The substrate's cosmological derivations across Chapters 1–11 produce many testable observables. SP-27 is the framework's structured program for systematic comparison of substrate predictions against observational data.

12.1 The SP-27 program

SP-27 (Casey directive May 16, 2026) is the Observational Reanalysis Program — a multi-year effort to reanalyze existing cosmological survey data (Planck, DES, eBOSS, etc.) against BST-substrate primary predictions, identifying where the substrate framework's predictions agree with observation, where they diverge, and where the divergences point to substrate-side refinement opportunities.

The current program scope (~30% complete) covers:

- CMB temperature + polarization spectra
- Supernovae Hubble-diagram analysis
- BAO scale measurements
- Cluster cosmology
- Weak-lensing surveys

12.2 Methodology

Per chapter analysis runs through:

1. Identify the substrate-derived prediction for the observable
2. Compare to observed value with reported uncertainties
3. Compute deviation in BST primary tier (D/I/C/S)
4. Flag for honest-scope refinement or D-tier ratification
5. Cross-reference to the catalog and the K-audit chain

12.3 Falsifier table

The chapter's payload is the per-observable falsifier table — what BST predicts, what would refute the prediction, current experimental status. Sample entries:

- H_0 : BST predicts ~ 67 -68 km/s/Mpc (Chapter 5, four routes). Refuted if SH0ES distance ladder systematic is resolved and gives > 70 km/s/Mpc.
- n_s : BST predicts $1 - 5/137 = 0.9635$. Refuted if Planck-precision measurement diverges by $> 1\%$.
- Λ matches to 0.076 dex. Refuted by precision improvement showing > 0.5 dex deviation.
- DM/baryon ratio: BST predicts 16/3. Refuted by precision improvement showing $> 2\%$ deviation.
- r (tensor-to-scalar): BST predicts ≈ 0 . Refuted by detection of $r > 0.01$.

12.4 What comes next

Volume 5 — Quantum Mechanics from D_IV⁵ — develops the substrate-derivation of quantum mechanics' foundations, building on the Born=Bergman result of Volume 1 Chapter 7 and treating the measurement problem, Bell inequalities, decoherence, and POVMs at substrate-mechanical depth.

The reader who has worked through Volume 4 has now seen the substrate's cosmological derivations end-to-end. The cosmological constant is resolved at 99.94%; the Hubble tension has a substrate-side resolution; CMB structure matches; BBN matches; the dark sector emerges from substrate Shannon channels. The substrate produces cosmology from the same five integers that produce particle physics.

Where to look this up: SP-27 program: Casey directive May 16, 2026; multi-year ongoing. For cosmological observation reviews: Planck 2018 results; DES Y3 release.